Warren Lin

Objective

Self-directed electrical engineering enthusiast with a passion for hardware repair, embedded systems, and digital restoration. I specialize in revitalizing "dead" or obsolete tech, such as installing Linux on 20-year-old computers, through custom firmware development, reverse engineering, and in-depth diagnostics. My goal is to apply hands-on experience and creative problem-solving skills in Physics Academia and Electrical Engineering with a focus on embedded circuits.

Education

2023–2024 **Linkou Kang Quo International School**, *Linkou, New Taipei City, Taiwan*
Clubs: Model United Nations, EE Club, Policy/Public Forums. Academic Interests: Electrical Engineering, Material Sciences, Policy Debate. Relevant Courses: Honors World History, Honors Band

2024–2026 **San Ramon Valley High School**, *Danville, California*
Clubs: Model United Nations, Team 1280 Robotics Team, IEEE Club. Academic Interests: Electrical Engineering, Quantum Information Theory, Applied Mathematics, Material Sciences, Policy Debate. Relevant Courses: AP Calculus AB, AP Computer Science A, Robotics, Auto Tech

Technical Contributions & Hacktivities

Business and Electrical Lead / FRC Robotics Team 1280

2024–Present **Ragin' C Biscuit**, *Danville, California*
Business: Write grant 50+ applications, email cooperate sponsors; manage and order parts, managed club funds of $50,000, built a project management system on Notion to ensure we meet cooprate deadlines and hit funding targets. I helped get 34% more than the previous fiscal year. **Electrical**: Train the 30 new members in basics and safety of electrical systems, provide hands-on experience; designed a system to detect and isolate CAN Bus errors, built a system for simple debugging and maintenance,

MIT Open Compute Lab

2025–Present **MIT OCL**, *Cambridge, Massachusetts*
Explored the philosophy of open-source hardware and the extent to which components can be considered open. I worked with graduate students, undergrad-students ranging from the East to the West coast. We discussed and communicated primarily online, contributions managed by Github. I audited many RISC-V confrences from online and presented findings and new extensions. Contributed to the open source CVA6 RISC-V core and helped develop a clone project. Actively working on various MIT OCL projects from the Tiny GPU to a Operating System that runs doom on RISC-V. Our eventual goal: Develop an SoC based on RISC-V architecture, comparable to modern systems like the Apple M1, with powerful GPU support and extensive Linux compatibility at a much lower cost,

Hardware Restoration

2019–Present **InLabs**, *Danville, California; Beitou, Taipei, Taiwan*
Reverse engineering and preserve hardware deemed "old." Installed Linux on old family computers, comparing performance with Windows and making hardware modifications to improve performance. I found that the linux kernel is much more efficient at handling multi-threaded, a truly efficient parallel computing operating system. Daily drove a 2012 laptop, teaching me to be creative with limited computing resources. Reverse-engineered game consoles to install Linux and helped friends jailbreak or reflash firmware on old devices for new use cases.

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Berkeley ML Research

Summer 2025 **CCE**, *Berkeley, California*

Developed a Reinforcement Learning model to predict stock outcomes. Created a training environment where the agent analyzes 15 stocks and predicts future performance. In Result, I've significantly overestimated the training data and created a strong weights that could determine stocks future trajectory. My Deep Q-Learning model has consistantly predicting high surges at rantom times accurately. Strengthened skills relevant to the current AI boom, focusing on learning over profit. Developed nuanced understanding of the AI Boom and strong belief in ethical AI development.

Creative Interests

Musical

KCIS Band Director: Tenor Saxophone, Alto Saxophone

Personal: Piano (2015–Present), Bass Guitar

STEM

Creative Engineering: Retro tech restoration, circuit analysis, custom keyboards, sound engineering

Open Tech: Open-source projects, open electronics, ethical hacking

Personal: Public discourse, problem-solving challenges, reading system-level documentation

Policy

Politics: Constant news intake, policy letters, political discourse

Diplomatic: Independent journalism, humanitarian support

Academic Enrichment

Electrical Engineering: Circuit diagnostics, soldering, board-level repair, component testing

Reverse Engineering: Firmware analysis, system patching, memory mapping, logic probing

Embedded Systems: Arduino, Raspberry Pi, custom MCU projects

Programming: Python, Julia, Rust, Fortran95, Cobol, C/C++, Bash/Shellscript, Java (intermediate), Assembly (x86, Arm64, RISC-V64), Verilog

Tools: Multimeter, Oscilloscope, Logic Analyzer, JTAG Probing, KiCad, Linux, LaTeX

Soft Skills: Detail-oriented, systems management, debate and public speaking, patience under pressure, music-trained discipline, self-taught learner

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